

DARSHAN GOWDA M

AI Engineer | GenAI & RAG Systems | LLM Applications | Cloud ML

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PROFESSIONAL SUMMARY

AI Engineer at Tata Consultancy Services with 2 years of experience building production GenAI systems, RAG pipelines, and agentic AI workflows across regulated domains — biopharmaceutical, aviation, and medical devices. Shipped end-to-end AI solutions using Azure OpenAI (GPT-4o), LangGraph, and vector databases across aviation, healthcare, and enterprise reporting domains, cutting manual effort and enabling data-driven decisions for enterprise clients. Focused on Responsible AI, governance, and scalable cloud ML deployment.

TECHNICAL SKILLS

- **GenAI & LLMs:** Azure OpenAI (GPT-4o), Cohere command-r-plus, RAG Pipelines, Agentic AI, LangGraph, LangChain
- **Prompt Engineering:** Extraction & generation prompts, temperature tuning, JSON-mode, few-shot prompting
- **Vector & Search:** Qdrant, FAISS, BM25Plus, TF-IDF, RapidFuzz, Hybrid Search, Cosine Similarity
- **ML & Statistics:** Random Forest, Voting Ensemble, Predictive Modelling, Feature Engineering, Scikit-learn
- **Cloud Platforms:** AWS (S3, SageMaker, boto3), Azure (Azure OpenAI, Azure AI Studio)
- **Languages & Dev:** Python, SQL, Java, FastAPI, Streamlit, Node.js, Express
- **Data & Visualization:** Pandas, NumPy, Plotly, Matplotlib, Power BI, Excel
- **PDF & Document AI:** PyMuPDF, PyPDF2, FPDF, ReportLab, langextract

WORK EXPERIENCE

Assistant System Engineer — AI & Data — Tata Consultancy Services *Aug 2024 – Present | Hyderabad, India*

Vectorless PageIndex RAG | Aviation Domain — Azure OpenAI GPT-4o, BM25Plus, langextract

- Developed an embedding-free retrieval pipeline for EASA Airworthiness Directive documents using LLM-generated hierarchical document trees with entity enrichment (aircraft models, ATA codes, action keywords) and BM25Plus re-ranking, improving retrieval precision over baseline keyword search.
- Applied 6 systematic improvements over baseline RAG — entity-enriched tree nodes, iterative retrieval with sufficiency checks, query decomposition, normalized metadata filtering, BM25 post-filtering, and context budget management.

Medical Device Regulatory Submission Assistant | Healthcare Domain — RAG, Qdrant, Cohere, FastAPI

- Architected a RAG pipeline (Qdrant, Cohere command-r-plus at temperature=0.0, FastAPI) to automate FDA 510(k) predicate discovery, eliminating an estimated 80% of manual document search effort across a corpus of cleared medical device PDFs.
- Extracted Intended Use, Technical, and Design Characteristics via a two-layer similarity system: dense vector cosine search for Intended Use matching and TF-IDF/RapidFuzz sentence-level matching for technical characteristics, auto-generating compliant 510(k) Summary Reports.

Agentic Report Generation System | Enterprise Reporting Domain — LangGraph, FAISS, BM25, Azure OpenAI GPT-4o

- Engineered a 5-agent LangGraph pipeline (Planner, Retrieval, Writer, Validation, Assembly) with conditional routing and self-correcting validation loops, achieving 2–3x faster throughput vs. sequential processing.
- Combined 70% FAISS dense-vector retrieval with 30% BM25 sparse keyword search, and parallelized report sections with asyncio/ThreadPoolExecutor; delivered via FastAPI and Streamlit for self-service use across 5+ business teams.
- Standardized LLM prompt engineering practices across projects (temperature=0.0 for extraction, temperature=0.2 for generation), reducing hallucination incidents and improving output reliability in regulated environments.

CERTIFICATIONS

AWS Partner: Generative AI Essentials • Microsoft Certified: Azure Fundamentals (AZ-900) • Microsoft Certified: Power Platform Fundamentals • Data Science Certification (Internshala)

EDUCATION

B.E., Electronics and Communication Engineering — Don Bosco Institute of Technology (DBIT), Bengaluru *Dec 2020 – May 2024*

ACHIEVEMENTS

- 7th place, PromptWars: Hyderabad — Google/hack2skill AI Hackathon (July 2026)
- 3rd place, Inter-Departmental Mini Project Exhibition (outperformed 25 teams)